The Cross Product

* The cross product or vector product is Vector, not a scalar.

* The cross product of 2 vectors, $\vec{u}$ and $\vec{v}$, is a vector that is $\perp$ to both $\vec{u}$ and $\vec{v}$. (Hence, it is only defined in 3-space).

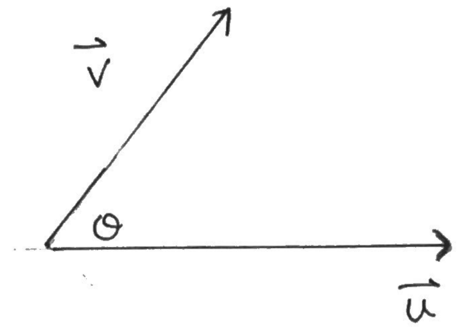
* Useful in angular velocity, statics & dynamics, area of triangles and parallelograms and volume of parallelepipeds.

* Notation:

$$\vec{u} \times \vec{v} = \text{cross product}$$

↑ cross, not multiply

* Direction of cross product is $\perp$ to $\vec{u}$ and $\vec{v}$ such that $\vec{u}$, $\vec{v}$ and $\vec{u} \times \vec{v}$ form a right-handed system.



$\vec{u} \times \vec{v}$

→ place your thumb on $\vec{u}$ and index finger on $\vec{v}$.

→ stick out your middle finger.

→ The direction of your middle finger is the direction of $\vec{u} \times \vec{v}$ (out of the page)

$\vec{v} \times \vec{u}$

→ place your thumb on $\vec{v}$ and index finger on $\vec{u}$.

→ stick out your middle finger.

→ The direction of your middle finger is the direction of $\vec{v} \times \vec{u}$ (into the page)

Properties of Cross Product

1. $\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$
2. $(\vec{u} + \vec{v}) \times \vec{w} = \vec{u} \times \vec{w} + \vec{v} \times \vec{w}$
3. $(k\vec{u}) \times \vec{v} = k(\vec{u} \times \vec{v}) = \vec{u} \times (k\vec{v})$, $k \in \mathbb{R}$.
4. $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$
5. Magnitude of $\vec{u} \times \vec{v}$ is $|\vec{u} \times \vec{v}| = |\vec{u}||\vec{v}|\sin(\theta)$, where θ is the angle between vectors.

Given $\vec{u} = [u_1, u_2, u_3]$ and $\vec{v} = [v_1, v_2, v_3]$

then $\vec{u} \times \vec{v} = [u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1]$

Example: Determine $\vec{u} \times \vec{v}$ if $\vec{u} = [2, -1, 4]$

and $\vec{v} = [-3, 7, -2]$

$$\begin{array}{cccccc} 2 & -1 & 4 & 2 & -1 & 4 \\ -3 & 7 & -2 & -3 & 7 & -2 \end{array}$$

$\therefore \vec{u} \times \vec{v} = [2, -1, 4] \times [-3, 7, -2]$

$= [-26, -8, 11]$

Check: Since $\vec{u} \times \vec{v}$ is $\perp$ to both $\vec{u}$ and $\vec{v}$

then $\vec{u} \times \vec{v} \bullet \vec{u}$ should equal 0 and

$\vec{u} \times \vec{v} \bullet \vec{v}$ should equal 0.

$\therefore [-26, -8, 11] \bullet [2, -1, 4] = -52 + 8 + 44 = 0$

$\therefore [-26, -8, 11] \bullet [-3, 7, -2] = 78 - 56 - 22 = 0$

$\Rightarrow \vec{u} \times \vec{v}$ is $\perp$ to both $\vec{u}$ and $\vec{v}$.

Example 1: If $|\vec{u}| = 8$, $|\vec{v}| = 5$ and the angle θ between $\vec{u}$ and $\vec{v}$ is $30°$, determine

a) $|\vec{u} \times \vec{v}|$

$= |\vec{u}||\vec{v}| \sin\theta$

$= (8)(5)\sin 30°$

$= 20$

b) $|\vec{v} \times \vec{u}|$

$= |\vec{v}||\vec{u}| \sin 30°$

$= (5)(8)\sin 30°$

$= 20$

Example 2: If $\vec{u} = [-1, 2, 1]$, $\vec{v} = [1, -1, -1]$ and $\vec{w} = [1, 1, -2]$, determine

a) $(\vec{u} \times \vec{v}) \bullet \vec{w}$

$= [-1, 0, -1] \bullet [1, 1, -2]$

$= -1 + 0 + 2$

$= 1$

note

$\vec{u} \times \vec{v} = [-1, 2, 1] \times [1, -1, -1]$

$$\begin{array}{cccccc} -1 & 2 & 1 & -1 & 2 & 1 \\ 1 & -1 & -1 & 1 & -1 & -1 \end{array}$$

$\vec{u} \times \vec{v} = [-1, 0, -1]$

b) $(\vec{u} \times \vec{v}) \times \vec{w}$

$= [-1, 0, -1] \times [1, 1, -2]$

$= [1, -3, -1]$

note:

$$\begin{array}{cccccc} -1 & 0 & -1 & -1 & 0 & -1 \\ & \times & \times & \times & & \\ 1 & 1 & -2 & 1 & 1 & -2 \end{array}$$

Notes:

① The product $(\vec{u} \times \vec{v}) \bullet \vec{w}$ is called the "triple scalar product" because it produces a scalar quantity.

② The product $\vec{u} \times \vec{v} \times \vec{w}$ is called the "triple vector product" because it produces a vector quantity.